VL-Rethinker: Incentivizing Self-Reflection of Vision-Language Models with Reinforcement Learning

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Abstract

Slow-thinking reasoning systems have substantially advanced text-only tasks, yet multimodal counterparts continue to lag behind. In this work, we propose `VL-Rethinker`, which incentivizes explicit self-reflection in vision-language models (VLMs) via reinforcement learning (RL), without relying on costly knowledge distillation. We identify a critical bottleneck in standard Group Relative Policy Optimization (GRPO): *vanishing advantages* caused by reward saturation, which starves the learning signal and discourages deeper reasoning. To address this, we introduce two complementary techniques. First, **Selective Sample Replay (SSR)** retains only non-zero-advantage samples from each rollout and prioritizes replay by advantage magnitude, yielding a curriculum-like RL that stabilizes training and reshapes the advantage distribution. Second, **Forced Rethinking** appends explicit self-verification trigger tokens to a fraction of rollouts, training the model to revisit and correct its reasoning without any teacher supervision. Together, these components form GRPO-SSR, which is near on-policy, uses binary outcome rewards, and is distillation-free. `VL-Rethinker` is trained on approximately 38.9K curated, verifiable multimodal queries spanning mathematics, science, and diverse disciplines. Experimental results demonstrate that `VL-Rethinker` achieves new open-source state-of-the-art performance across multiple multimodal reasoning benchmarks at both 7B and 72B scales. Notably, at 72B, `VL-Rethinker` attains 80.4% on MathVista testmini (+5.6 over prior open-source best), 63.5% on MathVerse testmini (+6.3), and 44.9% on MathVision test (+6.8). Our 7B model also surpasses prior 7B reasoning models by 2–6 absolute points across benchmarks, confirming that combining SSR-stabilized RL with training-time rethinking triggers effectively closes a key gap in multimodal slow-thinking.

1 Introduction

The emergence of slow-thinking reasoning systems—models that engage in extended chain-of-thought deliberation before producing answers—has driven remarkable progress in text-only tasks such as competition mathematics and scientific reasoning [Jaech et al., 2024, Guo et al., 2025, Team et al., 2025, Muennighoff et al., 2025]. These systems leverage reinforcement learning (RL) with outcome-based rewards to incentivize self-verification, backtracking, and reflection during reasoning. However, analogous capabilities in vision-language models (VLMs) remain substantially underdeveloped: while models such as OpenAI o1 excel in text-only domains, their multimodal counterparts exhibit limited capacity for genuine reflective reasoning over visual content [Hao et al., 2025, Yue et al., 2024a].

Several recent works have attempted to bridge this gap by applying RL directly to VLMs [Yang et al., 2025, Deng et al., 2025, Meng et al., 2025, Zhang et al., 2025, Huang et al., 2025, Peng et al., 2025], or by distilling long chain-of-thought (CoT) reasoning from text-only models into multimodal architectures [Du et al., 2025, Dong et al., 2024, Thawakar et al., 2025]. Despite these efforts, two fundamental challenges remain unresolved. First, standard GRPO [Guo et al., 2025]

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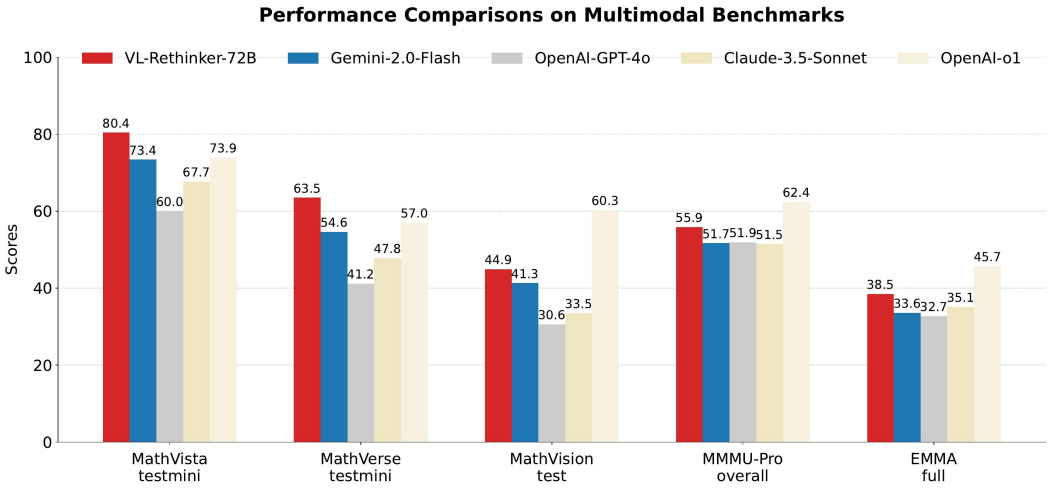

Figure 1: Performance comparison between `VL-Rethinker` and other SoTA models on different multimodal reasoning benchmarks. `VL-Rethinker` achieves new open-source SoTA at both 7B and 72B scales.

suffers from *vanishing advantages*: as the model improves, more and more rollouts either all succeed or all fail, producing zero group-relative advantage and providing no gradient signal. This instability undermines training and prevents models from learning sophisticated multi-step reasoning. Second, self-reflection behaviors such as self-checking, questioning, and correction are difficult to elicit via RL without explicit training-time encouragement, since they require the model to voluntarily revisit and revise its reasoning.

In this work, we propose `VL-Rethinker`, which addresses both challenges through two synergistic techniques. The core observation is that the vanishing advantages problem in GRPO is analogous to the *dead zone* problem in experience replay for deep RL [Schaul et al., 2015]: saturated batches yield near-zero gradients, wasting compute and stalling learning. To counter this, we introduce **Selective Sample Replay (SSR)**, which filters out zero-advantage samples from each training batch and replays high-advantage experiences proportional to their absolute advantage. This ensures the model always trains on informative decision-boundary cases where improvement is possible, creating a natural curriculum effect. To further promote self-verification during inference, we introduce **Forced Rethinking**: during a fraction q of rollouts, explicit rethinking trigger tokens are appended (e.g., phrases that prompt self-checking, questioning, or correction), and trajectories that lead to correct answers after rethinking are retained with an auxiliary SFT loss. This teaches the model to verify and revise its answers, rather than committing prematurely to a first-pass response.

Our method is trained on a carefully curated dataset of approximately 38.9K multimodal queries with verifiable answers, spanning math-focused domains, scientific reasoning, and general multi-discipline understanding. Using binary outcome rewards and a near on-policy update, `VL-Rethinker` does not require any teacher model or distilled reasoning traces.

Our contributions are as follows:

- We identify and formalize the *vanishing advantages* problem in GRPO for VLMs, explaining why standard RL training becomes unstable and produces shallow reasoning.
- We propose **Selective Sample Replay (SSR)**, which prioritizes high-advantage samples to stabilize RL training and create a curriculum-like learning dynamic, without discarding the near-on-policy guarantee.
- We introduce **Forced Rethinking**, a training-time mechanism that explicitly induces self-verification and correction in VLMs without distillation from teacher models.
- We demonstrate new open-source state-of-the-art performance on multiple multimodal reasoning benchmarks at 7B and 72B scales, surpassing previous open-source SoTA by substantial margins.

2 Background and Preliminaries

2.1 Vision-Language Models

Modern VLMs [Li et al., 2024, Bai et al., 2025, Chen et al., 2024a] typically consist of a visual encoder, a projector, and a large language model (LLM) backbone. The visual encoder processes image inputs into patch features, the projector maps these to the LLM embedding space, and the LLM generates responses conditioned on both the visual tokens and text tokens. Recent VLMs such as Qwen2.5-VL [Bai et al., 2025] and InternVL2.5 [Chen et al., 2024a] have demonstrated strong capabilities on visual understanding and reasoning tasks.

2.2 Reinforcement Learning for LLMs

Proximal Policy Optimization (PPO) [Schulman et al., 2017] and its variants are widely used for aligning LLMs with human preferences [Ouyang et al., 2022]. Group Relative Policy Optimization (GRPO) [Guo et al., 2025] is a simplified RL algorithm tailored for LLM reasoning, which eliminates the need for a separate critic by computing advantages from grouped rollouts. For a prompt x, GRPO samples n outputs $\{o_1, \ldots, o_n\}$ and computes the group-relative advantage for output o_i as:

$$\hat{A}_i = \frac{r_i - \text{mean}(\mathbf{r})}{\text{std}(\mathbf{r})}, \tag{1}$$

where r_i is the reward for output o_i and $\mathbf{r} = \{r_1, \ldots, r_n\}$. The GRPO objective is:

$$\mathcal{L}_{\text{GRPO}}(\theta) = -\mathbb{E}\left[\sum_{i=1}^{n} \min\left(\frac{\pi_\theta(o_i|x)}{\pi_{\theta_{\text{old}}}(o_i|x)}\hat{A}_i,\ \text{clip}\left(\frac{\pi_\theta(o_i|x)}{\pi_{\theta_{\text{old}}}(o_i|x)}, 1-\epsilon, 1+\epsilon\right)\hat{A}_i\right)\right]. \tag{2}$$

GRPO has been shown to effectively incentivize reasoning in LLMs [Guo et al., 2025, Zeng et al., 2025, Liu et al., 2025, Yu et al., 2025] and has been extended to multimodal settings [Meng et al., 2025, Yang et al., 2025, Deng et al., 2025].

2.3 Slow-Thinking in Vision-Language Models

Slow-thinking VLMs extend the deliberate reasoning paradigm to multimodal inputs. Prior approaches include SFT on distilled reasoning traces from GPT-4o or DeepSeek-R1 [Du et al., 2025, Dong et al., 2024, Thawakar et al., 2025, Yang et al., 2025], direct RL training on VLMs [Meng et al., 2025, Zhang et al., 2025, Deng et al., 2025], and hybrid SFT+RL pipelines [Huang et al., 2025, Peng et al., 2025]. The s1 work [Muennighoff et al., 2025] shows that appending "Wait" tokens during test time can improve text-only reasoning by forcing the model to reconsider. Our work is the first to integrate explicit rethinking triggers into the RL training loop for VLMs.

3 Method

3.1 The Vanishing Advantages Problem

A critical bottleneck in applying GRPO to VLMs is the *vanishing advantages* phenomenon. During training, a VLM progressively improves on easier queries, causing the reward distribution within each group to become increasingly homogeneous. Specifically, for binary rewards (correct/incorrect), if all n rollouts for a given prompt produce the same outcome (all correct or all incorrect), then $\hat{A}_i = 0$ for all i (see Eq. 1), yielding zero gradients. As illustrated in Figure 3, this occurs frequently with verifiable tasks: easy questions become trivially solved, while very hard questions consistently fail, leaving only a small fraction of "decision-boundary" questions with informative gradients.

This problem has two negative consequences. First, the effective batch utilization becomes very low, wasting compute on uninformative samples. Second, because binary-reward GRPO can only reinforce answers that differ in correctness, the model receives no signal to develop more sophisticated reasoning strategies such as backtracking or verification. The result is shallow, greedy reasoning rather than deliberate slow-thinking.

Figure 4 (left) shows that standard GRPO quickly develops a bimodal advantage distribution heavily concentrated near zero, confirming this analysis.

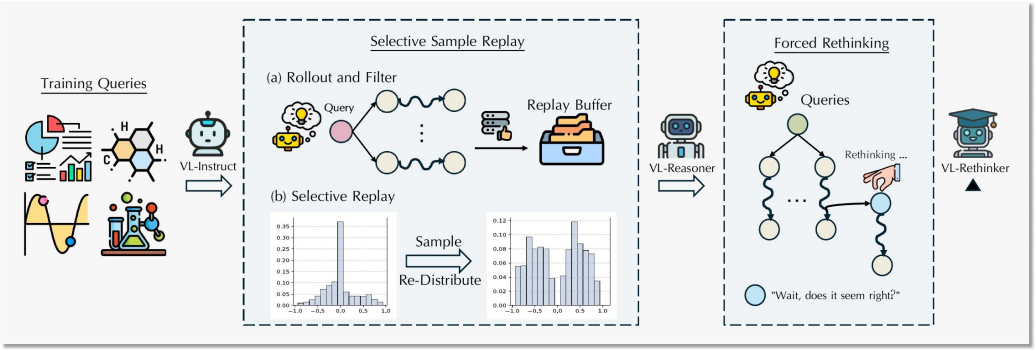

Figure 2: **Method Overview.** `VL-Rethinker` combines Selective Sample Replay (SSR) for stabilizing GRPO training with Forced Rethinking for incentivizing self-reflection. SSR filters zero-advantage samples and replays high-advantage experiences; Forced Rethinking appends rethinking triggers to a fraction of rollouts and retains successful rethinking trajectories with an auxiliary SFT loss.

3.2 Selective Sample Replay (SSR)

To address vanishing advantages, we propose **Selective Sample Replay (SSR)**, inspired by Prioritized Experience Replay [Schaul et al., 2015] in deep RL. The key idea is to maintain a rolling buffer of high-advantage experiences that can be replayed in future training batches, ensuring that the effective training distribution is always centered on informative decision-boundary cases.

Concretely, after each rollout batch, we:

1. **Filter**: Separate samples into non-zero-advantage samples $\mathcal{S}_+$ and zero-advantage samples $\mathcal{S}_0$.
2. **Store**: Add all samples from $\mathcal{S}_+$ to a persistent replay buffer $\mathcal{B}$, which retains the most recent high-advantage experiences across rollout rounds.
3. **Compose**: For each training step, compose the effective batch as a mixture of: (a) current non-zero-advantage samples from $\mathcal{S}_+$, and (b) replayed samples from $\mathcal{B}$, sampled with probability proportional to $|\hat{A}|^\alpha$ for a priority exponent α.

The composed batch thus always includes a sufficient number of informative samples, regardless of the current saturation level. Because the replay buffer is continuously updated with fresh rollouts, the method remains near-on-policy: replayed samples are typically only a few training steps old and still share similar policy characteristics with the current model. This differs from standard off-policy replay buffers used in value-based RL and is more analogous to the controlled shuffling in curriculum learning [Wang et al., 2022].

Figure 4 (right) shows that GRPO-SSR reshapes the advantage distribution, maintaining a healthy spread away from zero throughout training.

3.3 Forced Rethinking

Even with SSR stabilizing the RL optimization, the model may still learn to commit prematurely to first-pass answers without genuine self-verification. To explicitly train self-reflection, we introduce **Forced Rethinking**, a mechanism that encourages the model to revisit and potentially correct its reasoning.

During training, on a randomly selected fraction q of rollouts, we append a rethinking trigger phrase to the model's first-pass response. The trigger phrase takes various forms to encourage different self-reflection behaviors:

- *Self-checking*: "Wait, let me re-examine this..."
- *Correction*: "Actually, I think I made an error. Let me reconsider..."
- *Questioning*: "Hmm, is this really correct? Let me verify..."

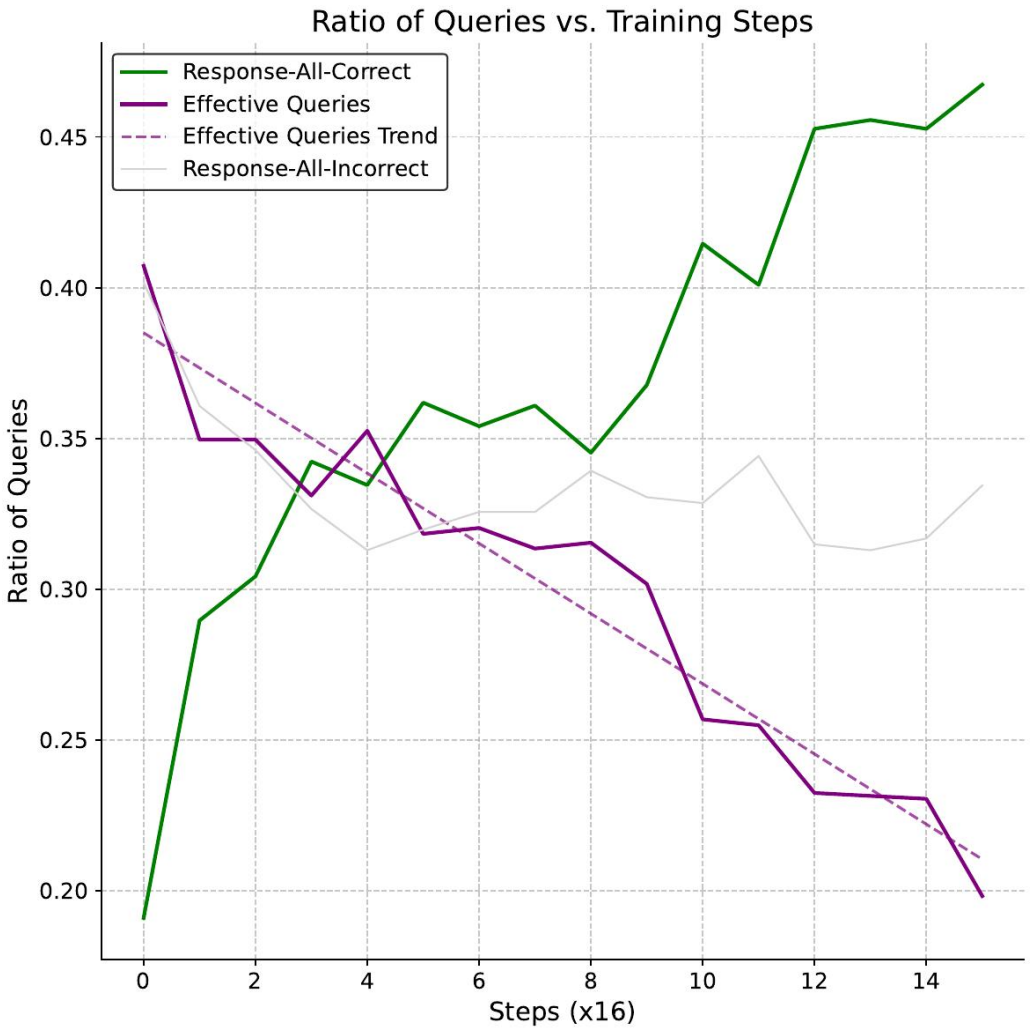

Figure 3: Illustration of the Vanishing Advantages problem. As training progresses, a growing fraction of prompts produce either all-correct or all-incorrect rollouts, resulting in near-zero group-relative advantages and providing no gradient signal.

After appending the trigger, the model continues generating an extended response. We then evaluate the correctness of the final answer in this extended response.

Trajectories where (a) the first-pass answer was wrong but the rethinking-extended answer is correct, are retained as high-quality self-correction demonstrations and used to compute an auxiliary SFT loss:

$$\mathcal{L}_{\text{SFT}}(\theta) = -\mathbb{E}\left[\log \pi_\theta(\text{rethinking response} \mid x, \text{trigger})\right], \tag{3}$$

applied only to the rethinking portion of the response. This loss teaches the model to correct errors rather than simply repeating them. The total training objective combines the GRPO-SSR loss with this auxiliary term:

$$\mathcal{L}(\theta) = \mathcal{L}_{\text{GRPO-SSR}}(\theta) + \lambda \mathcal{L}_{\text{SFT}}(\theta), \tag{4}$$

where λ is a coefficient controlling the auxiliary loss weight.

Importantly, Forced Rethinking differs from test-time compute scaling [Muennighoff et al., 2025] in that it is applied *during training*: the model learns when rethinking is beneficial rather than always extending its response at test time. This leads to more efficient inference, since the model learns to rethink selectively on questions where it is uncertain.

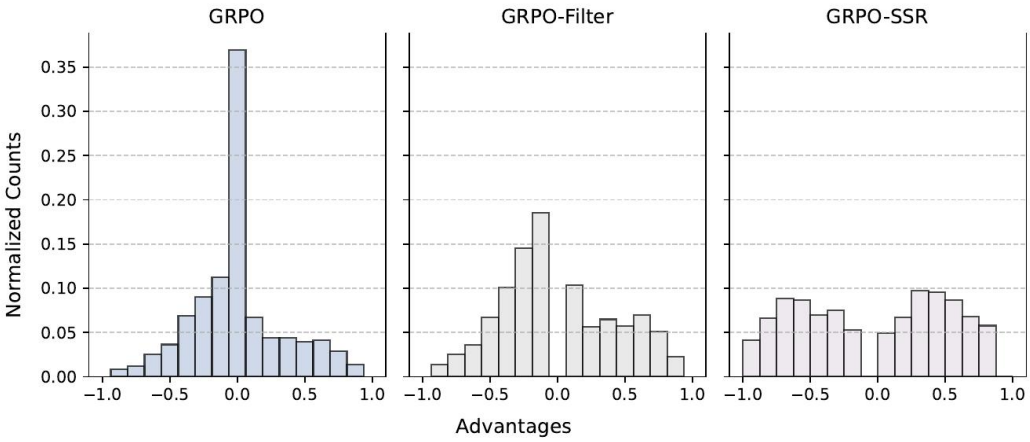

Figure 4: Comparisons of training dynamics of GRPO, GRPO-Filter and GRPO-SSR. SSR reshapes the advantage distribution away from zero, maintaining informative gradients throughout training.

Figure 5 illustrates the distinction between first-pass reasoning and rethinking-augmented reasoning, and Figure 6 compares different rethinking strategies.

3.4 Training Data

`VL-Rethinker` is trained on a curated dataset of approximately 38.9K multimodal queries with verifiable answers. The dataset is composed from three categories:

- **Math-focused** ($\sim$11K queries): Problems from mathematical reasoning benchmarks including geometry, algebra, and visual mathematics.
- **Science & Text** ($\sim$2K queries): Multi-discipline science problems requiring both textual and visual reasoning.
- **General multimodal** ($\sim$16K queries): Diverse visual question-answering tasks with verifiable answers spanning charts, diagrams, and real-world visual scenes. For the 7B model, we use a 16K subset; for the 72B model, we use the full 38.9K dataset.

All queries have binary-verifiable answers, enabling stable rule-based reward computation without a learned reward model.

3.5 Training Setup

`VL-Rethinker` is initialized from Qwen2.5-VL-Instruct [Bai et al., 2025] at 7B and 72B scales. We also present a 32B variant. Training uses the GRPO-SSR objective with the following hyperparameters: $n = 8$ rollouts per prompt, learning rate 10×10^{-7}, auxiliary SFT coefficient $\lambda = 0.05$, rethinking fraction $q = 0.2$, and maximum output length 6144 tokens. We use binary rewards: $r = 1$ for correct answers, $r = 0$ otherwise, with format compliance also rewarded. Training runs for 3 episodes over the dataset. The framework is built on OpenRLHF [Ouyang et al., 2022] with Ray-based distributed training.

4 Experiments

4.1 Benchmarks and Baselines

We evaluate on seven diverse multimodal reasoning benchmarks:

- **Math-related**: MathVista testmini [Lu et al., 2023], MathVerse testmini [Zhang et al., 2024], MathVision test [Wang et al., 2024a]

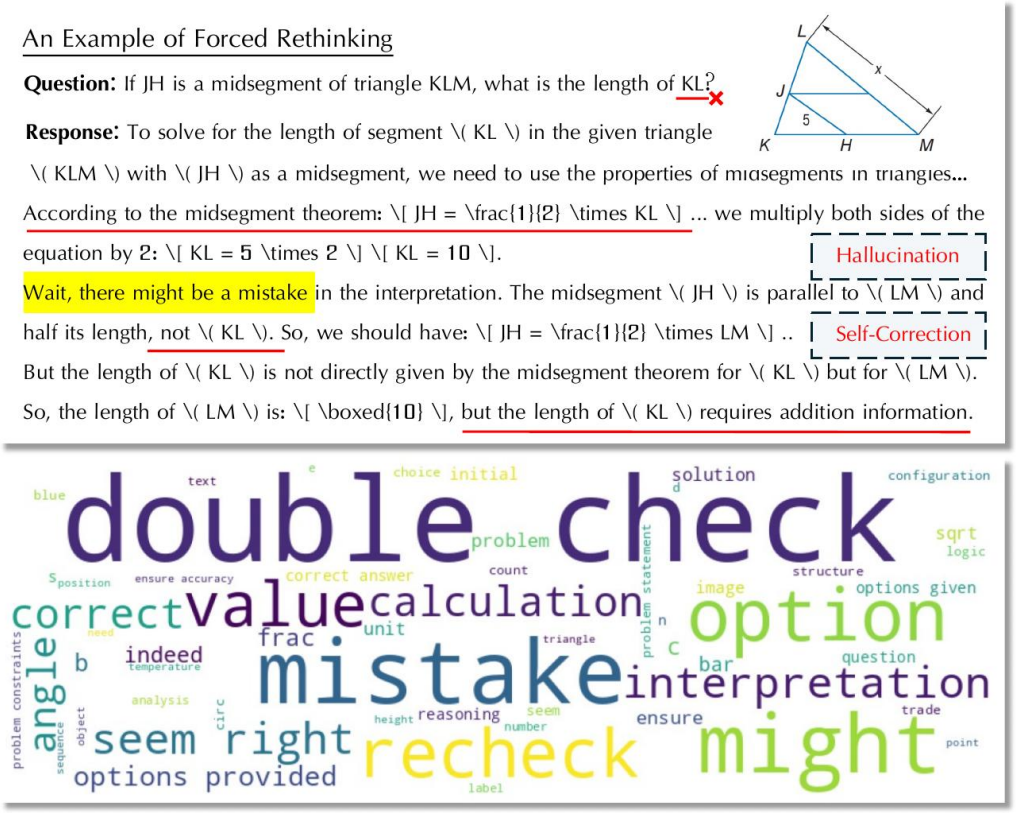

Figure 5: Illustration of the Vanishing Advantages problem and how Forced Rethinking addresses it by extending the reasoning chain with explicit self-verification.

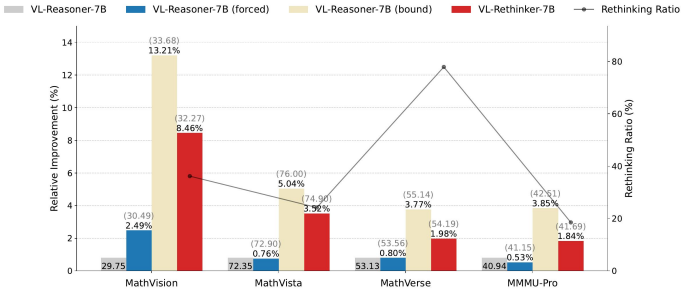

Figure 6: **Relative Improvement with Different Re-thinking Strategies.** Forced Rethinking with diverse trigger types yields the most consistent improvements across benchmarks.

- **Multi-discipline**: MMMU-Pro overall [Yue et al., 2024a], MMMU val [Yue et al., 2024b], EMMA full [Hao et al., 2025]

- **Real-world**: MEGA core [Chen et al., 2024b]

We compare against both proprietary models (OpenAI-o1 [Jaech et al., 2024], GPT-4o [Hurst et al., 2024], Claude-3.5-Sonnet [Anthropic, 2024], Gemini-2.0-Flash [Team et al., 2024]) and open-source baselines including general VLMs (InternVL2.5 [Chen et al., 2024a], Qwen2.5-VL [Bai et al., 2025], LLaVA-OV [Li et al., 2024]) and reasoning VLMs (MM-Eureka [Meng et al., 2025], R1-VL [Zhang et al., 2025], R1-Onevision [Yang et al., 2025], OpenVLThinker [Deng et al., 2025]).

Model	Math-Related			Multi-Discipline			Real-World
	MathVista testmini	MathVerse testmini	MathVision test	MMMU-Pro overall	MMMU val	EMMA full	MEGA core
Proprietary Model							
OpenAI-o1	73.9	57.0	60.3	62.4	78.2	45.7	56.2
OpenAI-GPT-4o	60.0	41.2	30.6	51.9	69.1	32.7	52.7
Claude-3.5-Sonnet	67.7	47.8	33.5	51.5	68.3	35.1	52.3
Gemini-2.0-Flash	73.4	54.6	41.3	51.7	70.7	33.6	54.1
Open-Source Models							
Llama4-Scout-109B	70.7	-	-	52.2	69.4	24.6	31.8
InternVL-2.5-78B	72.3	51.7	34.9	48.6	61.8	27.1	44.1
QvQ-72B	71.4	48.6	35.9	51.5	70.3	32.0	8.8
LLava-OV-72B	67.5	39.1	30.1	31.0	56.8	23.8	29.7
Qwen-2.5-VL-32B	74.7	48.5	38.4	49.5	†59.4	31.1	13.3
Qwen-2.5-VL-72B	74.8	57.2	38.1	51.6	†67.0	34.1	49.0
`VL-Rethinker`-32B	78.8	56.9	40.5	50.6	65.6	37.9	19.9
`VL-Rethinker`-72B	**80.4**	**63.5**	**44.9**	**55.9**	**68.8**	**38.5**	**51.3**
Δ (Ours - Open SoTA)	+5.6	+6.3	+6.8	+3.7	-1.4	+4.4	+2.3

Table 1: Comparison between our 72B model and other state-of-the-art models. The notation of † indicates reproduced results using our evaluation protocols.

4.2 Main Results at 72B Scale

Table 1 shows results at 72B scale. `VL-Rethinker`-72B achieves new open-source state-of-the-art performance across all benchmarks:

- **Math-related**: MathVista 80.4% (+5.6 over open-source SoTA), MathVerse 63.5% (+6.3), MathVision 44.9% (+6.8).
- **Multi-discipline**: MMMU-Pro 55.9% (+3.7), EMMA 38.5% (+4.4).
- **Real-world**: MEGA core 51.3% (+2.3).
- **MMMU val**: 68.8% (–1.4 compared to best open-source; Qwen2.5-VL-72B achieves 67.0% under our evaluation protocol).

Notably, `VL-Rethinker`-72B surpasses Gemini-2.0-Flash and approaches OpenAI-o1 on several benchmarks, despite being trained without proprietary data or architectural changes. We also present `VL-Rethinker`-32B results, which achieve competitive performance between the 7B and 72B variants.

Figure 7 shows training dynamics, confirming that GRPO-SSR produces a more stable and monotonically improving evaluation curve compared to standard GRPO.

4.3 Main Results at 7B Scale

Table 2 compares `VL-Rethinker`-7B against a comprehensive set of 7B-scale models. `VL-Rethinker`-7B achieves:

- MathVista 74.9% (+4.7 over OpenVLThinker-7B), MathVerse 54.2% (+6.3), MathVision 32.3% (+2.4).
- MMMU-Pro 41.7% (+4.4), MMMU val 56.7% (+0.7), EMMA 29.7% (+3.1), MEGA 37.2% (+2.2).

These results confirm that the proposed methods generalize across model scales, and that GRPO-SSR combined with Forced Rethinking provides consistent gains over purely RL-trained baselines such as MM-Eureka and R1-VL.

4.4 Ablation Study

Table 3 reports ablation results on the 7B model. We analyze the contributions of Forced Rethinking, SSR, and the training data composition.

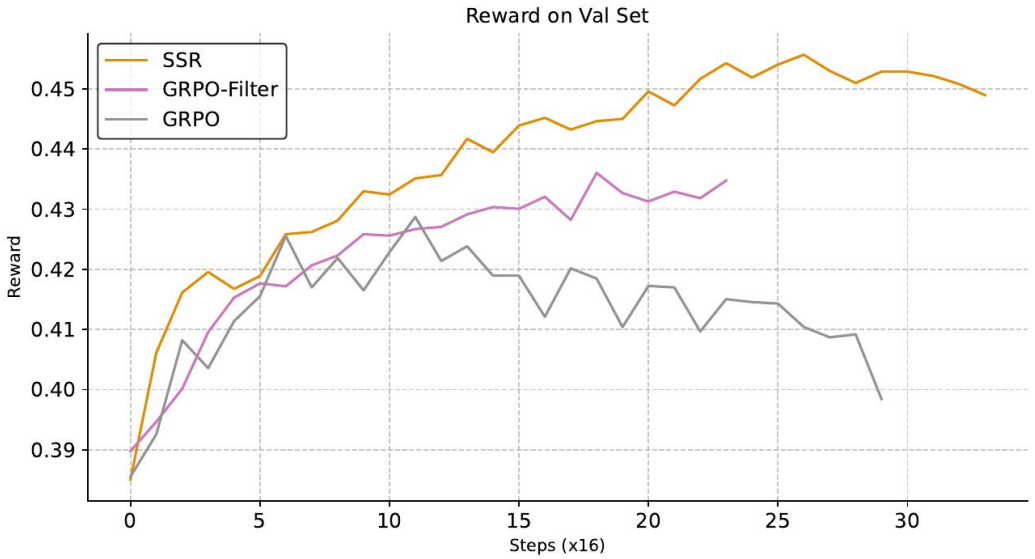

Figure 7: Comparisons of training dynamics of GRPO, GRPO-Filter and GRPO-SSR on evaluation performance. GRPO-SSR achieves more stable improvement throughout training.

Model	Math-Related			Multi-Discipline			Real-World
	MathVista testmini	MathVerse testmini	MathVision test	MMMU-Pro overall	MMMU val	EMMA full	MEGA core
General Vision-Language Models							
InternVL2-8B	58.3	-	17.4	29.0	51.2	19.8	26.0
InternVL2.5-8B	64.4	39.5	19.7	34.3	56.0	-	30.4
QwenVL2-7B	58.2	-	16.3	30.5	54.1	20.2	34.8
QwenVL2.5-7B	68.2	46.3	25.1	36.9	†54.3	21.5	35.0
Llava-OV-7B	63.2	26.2	-	24.1	48.8	18.3	22.9
Kimi-VL-16B	68.7	44.9	21.4	-	†55.7	-	-
Vision-Language Reasoning Models							
MM-Eureka-8B (Intern)	67.1	40.4	22.2	27.8	49.2	-	-
MM-Eureka-7B (Qwen)	73.0	50.3	26.9	-	-	-	-
R1-VL-7B	63.5	40.0	24.7	7.8	44.5	8.3	29.9
R1-Onevision-7B	64.1	46.4	29.9	21.6	-	20.8	27.1
OpenVLThinker-7B	70.2	47.9	25.3	37.3	52.5	26.6	12.0
VL-Rethinker-7B	**74.9**	**54.2**	**32.3**	**41.7**	**56.7**	**29.7**	**37.2**
Δ (Ours - Prev SoTA)	+4.7	+6.3	+2.4	+4.4	+0.7	+3.1	+2.2

Table 2: Comparison between our 7B model and other general and reasoning vision-language models. † means that the results are reproduced by us.

Effect of Forced Rethinking. Removing Forced Rethinking ("w/o 'Forced-Rethinking"') degrades performance across all benchmarks, most notably on MathVision (32.3% → 29.8%) and MathVista (74.9% → 72.4%), confirming that explicit self-verification training is a key contributor to the method's success.

Effect of SSR. Replacing SSR with simple filtering ("- no SSR") causes a further substantial drop on all metrics (e.g., MathVision: 29.8% → 28.5%, MathVerse: 53.2% → 50.0%). Using standard GRPO with neither SSR nor filtering ("- no SSR & Filter") yields the worst results (MathVision: 26.0%), confirming that both the filtering mechanism and the replay strategy are essential.

These results empirically validate the theoretical motivation in Section 3.1: vanishing advantages in standard GRPO directly harm model performance, and SSR effectively addresses this by ensuring informative gradients throughout training.

Model	RL-Algo	Data	MathVision	MathVista	MathVerse	MMMU-Pro	EMMA
VL-Rethinker-7B	SSR	16K	32.3	74.9	54.2	41.7	29.7
w/o 'Forced-Rethinking'	SSR	16K	29.8	72.4	53.2	40.9	29.5
- no SSR	Filter	16K	28.5	72.0	50.0	40.0	26.9
- no SSR& Filter	GRPO	16K	26.0	70.9	51.4	38.8	26.2
- no Text	SSR	13K	29.1	73.5	53.5	41.1	28.7
- no Science&Text	SSR	11K	28.0	71.6	50.3	39.7	28.0

Table 3: Ablation Results to show the impact of SSR and Data Mix.

Effect of Data Composition. Ablating data categories reveals that both math-focused and science/-text data contribute to performance. Removing text-only math data ("- no Text") drops MathVision from 32.3% to 29.1%, while removing both science and text data ("- no Science&Text") further reduces performance to 28.0%, particularly on multi-discipline benchmarks.

5 Related Work

Vision-Language Models. Large VLMs have achieved remarkable progress in visual understanding and reasoning. General VLMs such as LLaVA-OneVision [Li et al., 2024], InternVL2.5 [Chen et al., 2024a], Qwen2-VL [Wang et al., 2024b], and Qwen2.5-VL [Bai et al., 2025] excel on perception-heavy benchmarks but show limited capacity for complex multi-step reasoning. Our work targets specifically the reasoning deficit of VLMs.

Reasoning in LLMs and VLMs. Chain-of-thought prompting [Wei et al., 2022] and its extensions [Feng et al., 2023] demonstrated that explicit intermediate steps greatly enhance LLM reasoning. Building on this, works such as DeepSeek-R1 [Guo et al., 2025] and Kimi k1.5 [Team et al., 2025] showed that RL with simple outcome rewards can elicit long-chain reasoning in LLMs without SFT warm-up. For VLMs, slow-thinking has been explored via SFT on distilled reasoning data [Du et al., 2025, Dong et al., 2024, Thawakar et al., 2025, Xu et al., 2025], hybrid SFT+RL pipelines [Huang et al., 2025, Peng et al., 2025, Wen et al., 2025], and direct RL on VLMs [Meng et al., 2025, Zhang et al., 2025, Yang et al., 2025, Deng et al., 2025]. Our work differs in that we identify and directly address vanishing advantages, a fundamental training instability, and introduce explicit training-time rethinking triggers.

Curriculum and Prioritized Learning. Prioritized Experience Replay [Schaul et al., 2015] showed that replaying high-TD-error transitions improves sample efficiency in value-based RL. Curriculum learning [Wang et al., 2022] organizes training examples by difficulty to improve learning efficiency. SSR draws inspiration from both: it prioritizes high-advantage experiences (analogous to PER) and maintains a difficulty-aware training distribution (analogous to curriculum learning), but in the context of near-on-policy LLM fine-tuning.

Self-Verification and Reflection in LLMs. Process reward models [Lightman et al., 2023] provide step-level feedback to improve reasoning quality. Budget forcing [Muennighoff et al., 2025] uses "Wait" token insertion at test time to extend reasoning. Dr. GRPO [Liu et al., 2025] identifies optimization biases in GRPO and proposes unbiased corrections. DAPO [Yu et al., 2025] introduces decoupled clipping and dynamic sampling. Our Forced Rethinking is distinct in that it integrates self-reflection triggers directly into the RL training loop, training the model to selectively rethink rather than always extending its response.

6 Conclusion

We presented VL-Rethinker, a VLM training framework that combines Selective Sample Replay (SSR) and Forced Rethinking to incentivize explicit self-reflection via reinforcement learning. By identifying and addressing vanishing advantages in GRPO, SSR ensures stable, informative training throughout the learning process. Forced Rethinking complements this by explicitly training self-verification without relying on teacher distillation. Trained on $\sim$38.9K curated verifiable multimodal

queries, `VL-Rethinker` achieves new open-source state-of-the-art results at both 7B and 72B scales across diverse multimodal reasoning benchmarks. Our work demonstrates that closing the multimodal slow-thinking gap does not require proprietary data or complex architectures—it requires addressing the fundamental optimization challenges in VLM RL training.

Limitations. Our binary reward function may be insufficient for open-ended generation tasks where correctness is not easily verified. Forced Rethinking introduces additional training complexity via the auxiliary SFT objective, requiring careful hyperparameter tuning. Future work may explore richer reward signals, verifier-augmented training, and extending the approach to video understanding.

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